

JEE MAIN REVISION MASTER: ATOMIC STRUCTURE 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Quantum Numbers & Electronic Configuration	12 PYQs	Quantum Numbers
Standard	Heisenberg Uncertainty & De Broglie Wavelength	2 PYQs	Wave Mechanics
Standard	Bohr's Orbitals & Electronic Transitions	0 PYQs	Bohr Model

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Quantum Numbers & Electronic Configuration	$\text{Radial nodes} = n - l - 1$, $\text{Angular nodes} = l$, $\text{Total nodes} = n - 1$, $\mu_s = \sqrt{n(n+2)} \text{ BM}$
Heisenberg Uncertainty & De Broglie Wavelength	$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$, $\Delta x \cdot m \Delta v \geq \frac{h}{4\pi}$, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$
Bohr's Orbitals & Electronic Transitions	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$, $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Quantum Numbers & Electronic Configuration (High Yield — 12 PYQs)

Core Principle: Orbital shapes, quantum numbers (n, l, m_l, m_s), nodal surfaces, and anomalous configurations (Cr: $3d^5 4s^1$, Cu: $3d^{10} 4s^1$).

■ **Caution / JEE Pitfall:** Spin-only magnetic moment $\mu = \sqrt{n(n+2)}$ BM depends on n , which is the number of UNPAIRED electrons, not total electrons!

■ **Speed Trick:** Degeneracy of hydrogen atom at energy level n is n^2 (without spin) or $2n^2$ (including spin).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Which of the following is the CORRECT statement about Ψ^2 ?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following represents de Broglie equation?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of orbitals associated with 'N' shell of an atom is..." [Ans: N/A]

3.2 Heisenberg Uncertainty & De Broglie Wavelength (Standard Yield — 2 PYQs)

Core Principle: Limits of simultaneous precision in position and momentum, and dual wave nature of micro-particles.

■ **Caution / JEE Pitfall:** If uncertainty in position and momentum are equal ($\Delta x = \Delta p$), then $\Delta v = (1/2m)\sqrt{(h/\pi)}$.

■ **Speed Trick:** De Broglie wavelength of an electron accelerated through potential V volts: $\lambda \approx \frac{12.27}{\sqrt{V}}$ Å.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Which of the following represents de Broglie equation?..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "According to the Heisenberg's Uncertainty principle, the value of $\Delta b \cdot \Delta x$ for an object whose mass is 10^{-6} kg is $(h=6.626 \times 10^{-34})$..." [Ans: N/A]

3.3 Bohr's Orbitals & Electronic Transitions (Standard Yield — 0 PYQs)

Core Principle: Electronic energy levels in hydrogenic species, wave number $\bar{\nu}$, and emission spectral series.

■ **Caution / JEE Pitfall:** Rydberg formula requires multiplication by Z^2 for single-electron ions like He^+ ($Z=2$) and Li^{2+} ($Z=3$).

■ **Speed Trick:** Limiting line (series limit) of any spectral series occurs when $n_2 = \infty$ implies $\bar{\nu}_{\text{limit}} = (R Z^2 / n_1^2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Which of the following is the CORRECT statement about Ψ^2 ?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following represents de Broglie equation?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Which of the following is the CORRECT statement about Ψ^2 ?

- (A) Ψ^2 represents atomic orbit
- (B) Probability density of the electron at that point
- (C) $\Psi^2 \neq 0$ for nodes
- (D) Ψ^2 has no physical meaning

Correct Answer: None

Solution Analysis: The correct statement is: Option B: Probability density of the electron at that point. In quantum mechanics, Ψ is the wave function of an electron. The wave function itself, Ψ , does not have direct physical meaning. But the square of the wave function gives the probability density of finding the electron at a particular point. More correctly, we write it as: $|\Psi|^2$. For real wave functions, this is often written simply as: Ψ^2 . So, Ψ^2 tells us the probability density of the electron at that point. Also, at nodes, the probability of finding the electron is zero, so: